

Department of Computer Science

SUBMITTED TO: Sir Uzair

COURSE: Operating System Lab

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REGISTRATION NO# 242201009

SEMESTER: BSCS - 5

SECTION: A

ASSIGNMENT # 01

Date: 30/10/2026

Assignment Title:
Windows System Administration & Task Automation Challenge

Task 1 – Windows & Keyboard Shortcuts

Objective

To demonstrate useful Windows keyboard shortcuts that improve productivity and help system administrators perform common tasks efficiently.

Keyboard Shortcuts and Their Functions

No.	Shortcut Keys	Purpose
1	Win + E	Opens File Explorer to access files and folders quickly.
2	Win + R	Opens the Run dialog box to execute programs and commands.
3	Win + I	Opens Windows Settings.
4	Win + D	Minimizes all open windows and displays the desktop.
5	Win + L	Locks the computer to protect it from unauthorized access.
6	Win + X	Opens the Quick Link menu containing important system administration tools.
7	Ctrl + Shift + Esc	Opens Task Manager directly to monitor running applications and processes.
8	Alt + Tab	Switches between currently open applications.
9	Ctrl + Shift + N	Creates a new folder in File Explorer.
10	Win + Shift + S	Opens the screen snipping tool to capture a selected area of the screen.
11	Win + Pause/Break	Opens the system information page on Windows versions where this shortcut is supported.
12	Alt + F4	Closes the currently active window or application.

Practical Demonstration

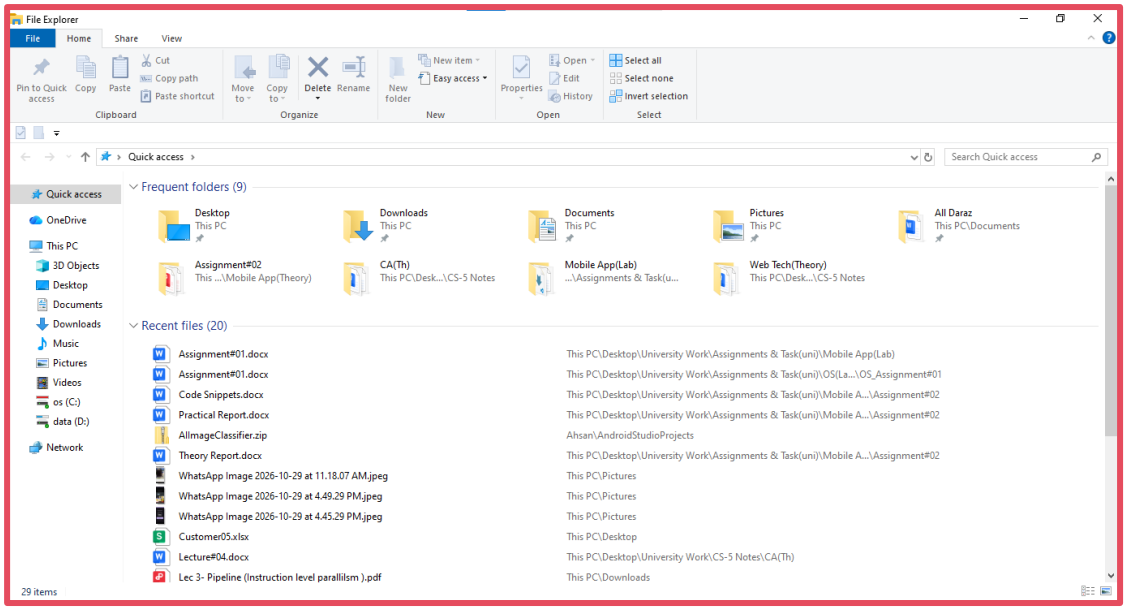
I practically tested these keyboard shortcuts on my Windows computer and observed their functions.

Examples of Practical Demonstration:

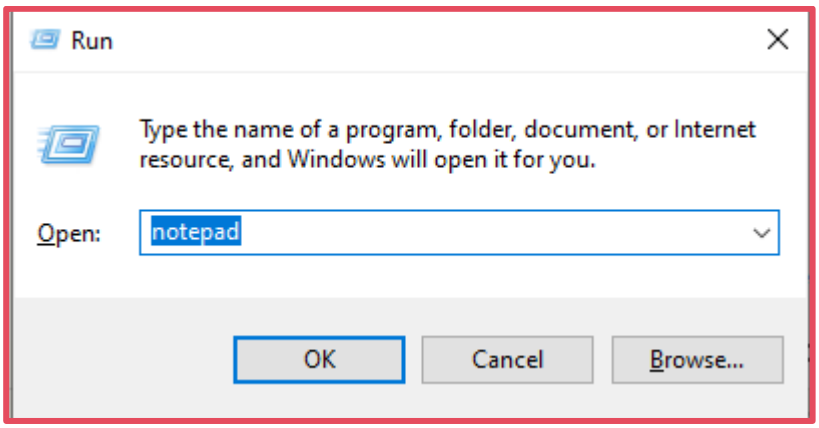
1. Pressed Win + E to open File Explorer.
2. Pressed Win + R to open the Run dialog box.
3. Pressed Win + I to access Windows Settings.
4. Pressed Ctrl + Shift + Esc to open Task Manager.
5. Pressed Win + X to access system administration options.
6. Pressed Win + Shift + S to capture a screenshot.

Screenshots to Insert:

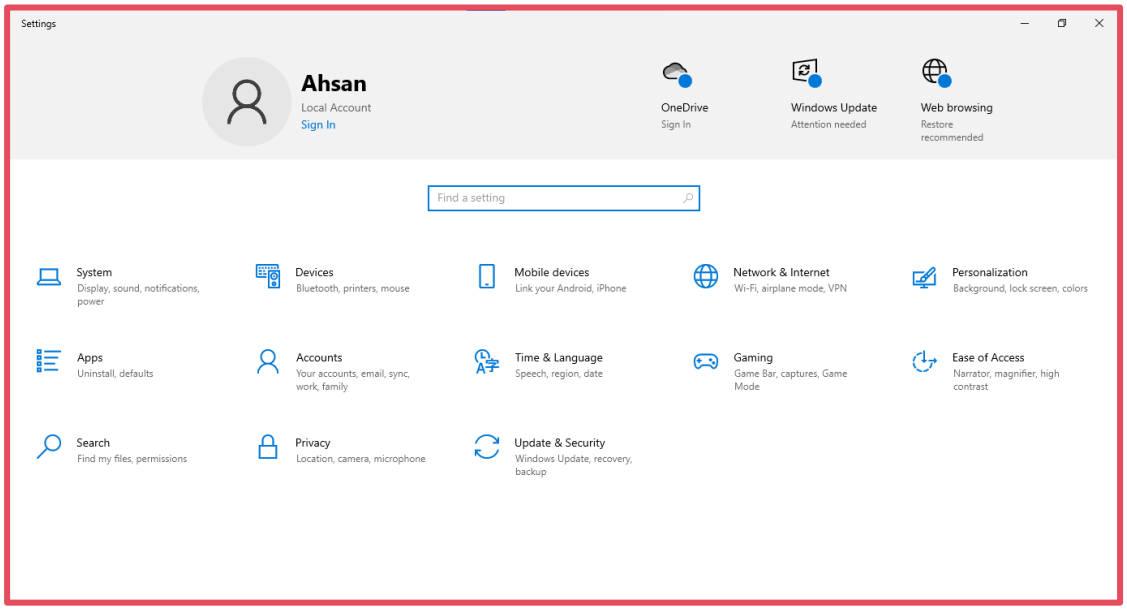
- **Screenshot 1: File Explorer opened using Win + E.**



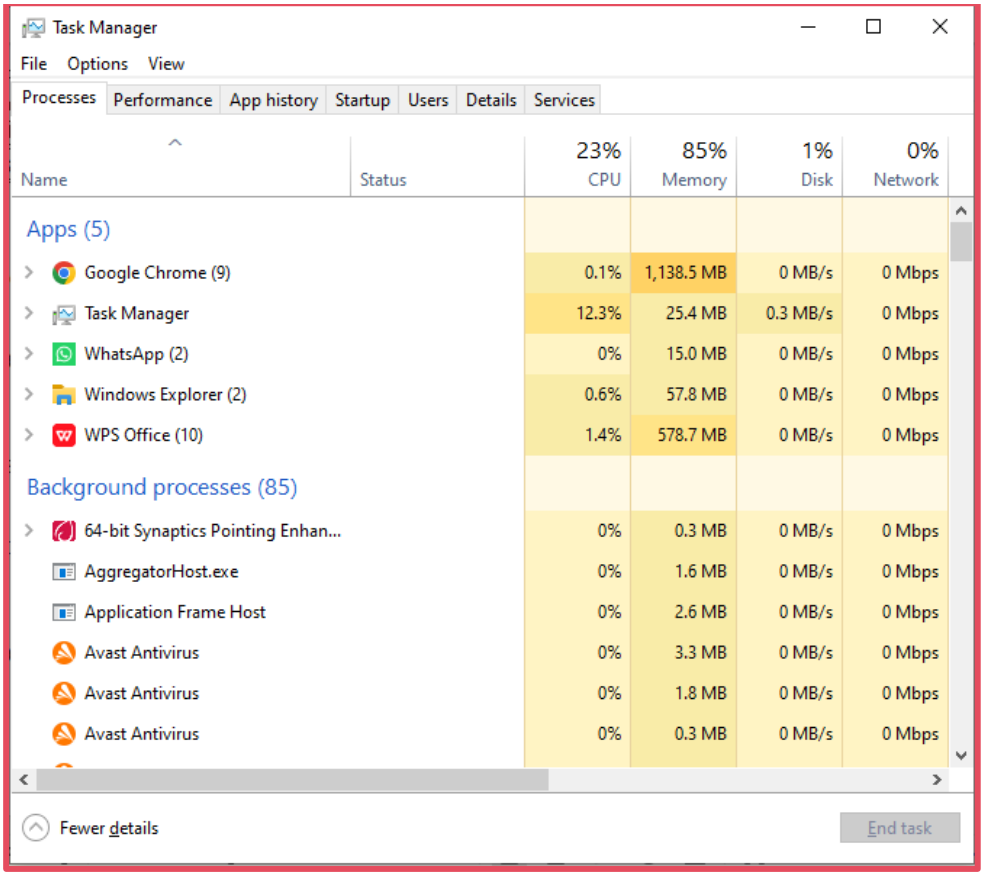
❖ Screenshot 2: Run dialog box opened using Win + R.



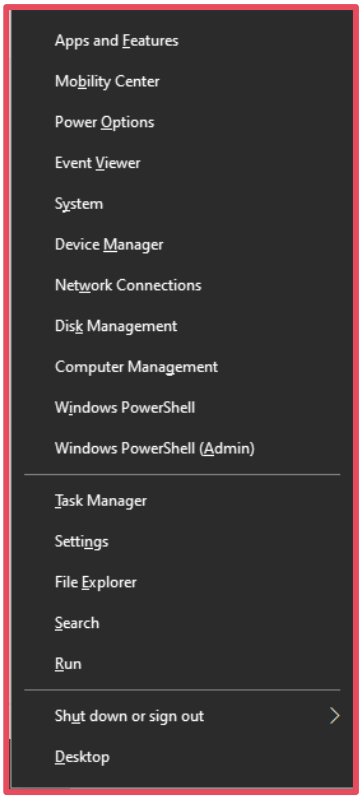
❖ Screenshot 3: Windows Settings opened using Win + I.



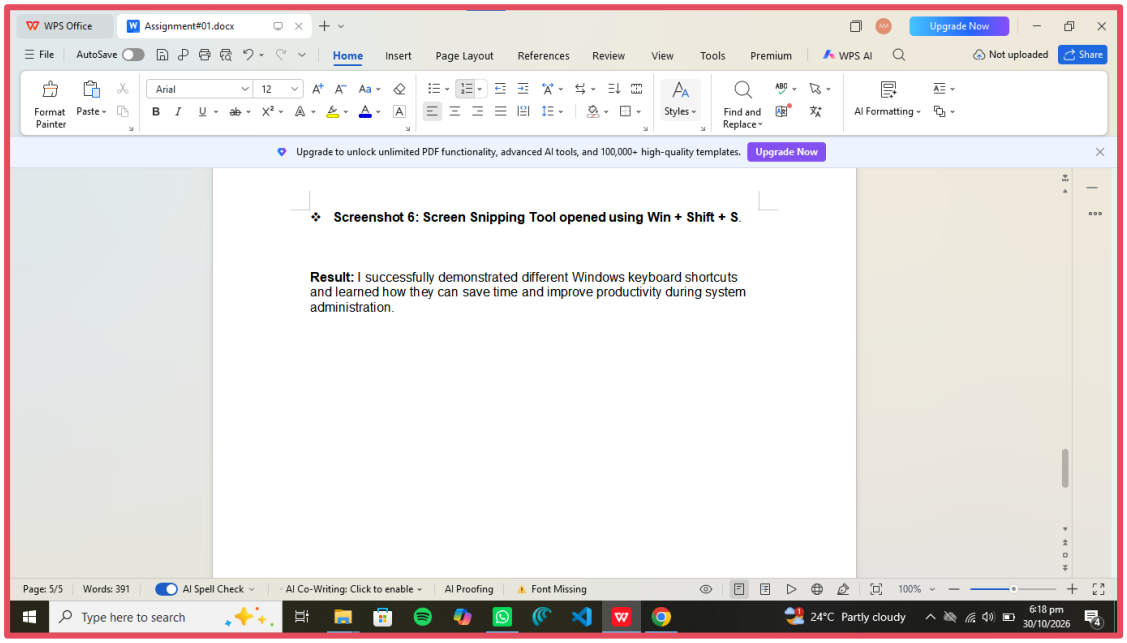
❖ **Screenshot 4: Task Manager opened using Ctrl + Shift + Esc.**



❖ **Screenshot 5: Quick Link menu opened using Win + X.**



❖ **Screenshot 6: Screen Snipping Tool opened using Win + Shift + S.**



Result: I successfully demonstrated different Windows keyboard shortcuts and learned how they can save time and improve productivity during system administration.

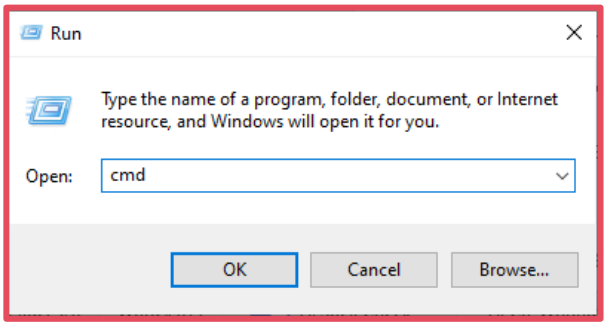
Task 2 – File & Directory Management Using CMD

Objective

To create and manage directories and text files using Command Prompt without using File Explorer.

Step 1: Open Command Prompt

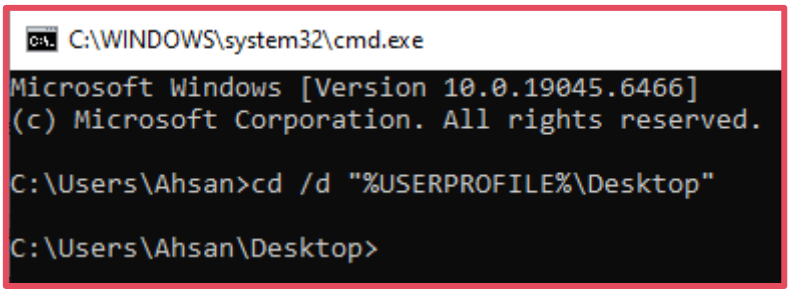
Press Win + R, type cmd, and press Enter to open Command Prompt.



Step 2: Create the Main Directory

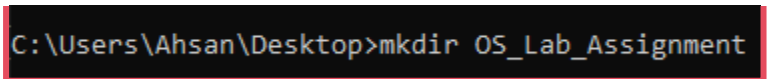
First, navigate to the Desktop:

cd /d "%USERPROFILE%\Desktop"



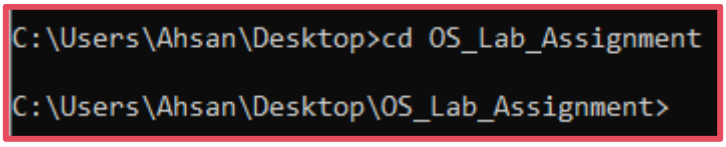
Create the main assignment directory:

mkdir OS_Lab_Assignment



Navigate into the newly created directory:

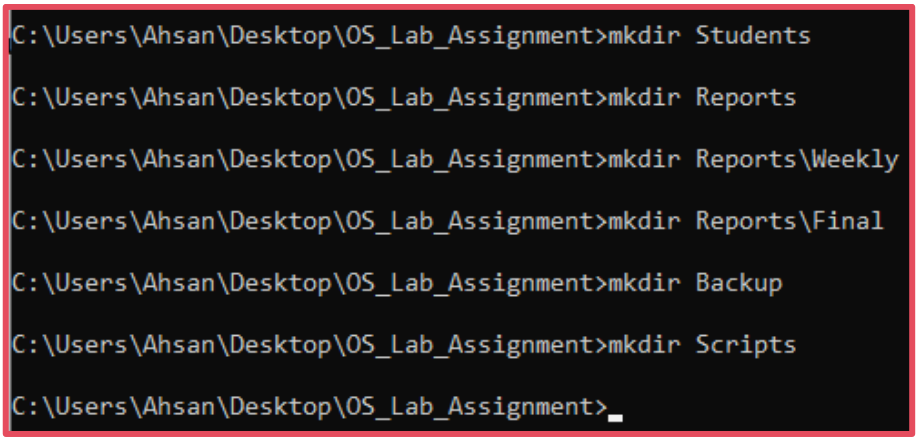
cd OS_Lab_Assignment



Step 3: Create the Required Directory Structure

Execute the following commands:

mkdir Students
mkdir Reports
mkdir Reports\Weekly
mkdir Reports\Final
mkdir Backup
mkdir Scripts



The following directory structure has now been created:

```
OS_Lab_Assignment
|
├── Students
├── Reports
|   ├── Weekly
|   └── Final
├── Backup
└── Scripts
```

Step 4: Display the Current Directory

- **Command:** ***cd***

```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>cd
C:\Users\Ahsan\Desktop\OS_Lab_Assignment
```

- **Purpose:** Displays the current working directory in Command Prompt.

Step 5: Display the Complete Directory Structure

- **Command:** ***tree***

```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>tree
Folder PATH listing for volume os
Volume serial number is BE2A-8502
C:.
├───Backup
├───Reports
│   ├───Final
│   └───Weekly
├───Scripts
└───Students
```

- **Purpose**: Displays the directory structure in a tree-like format.

Step 6: Create the Student Information File

Create a text file named student_info.txt and write personal academic information into it.

- **Commands:**

echo Name: Ahsan > student_info.txt
echo Registration Number: 242201009 >> student_info.txt
echo Semester: 5th Semester >> student_info.txt
echo Section: BSCS-5A >> student_info.txt

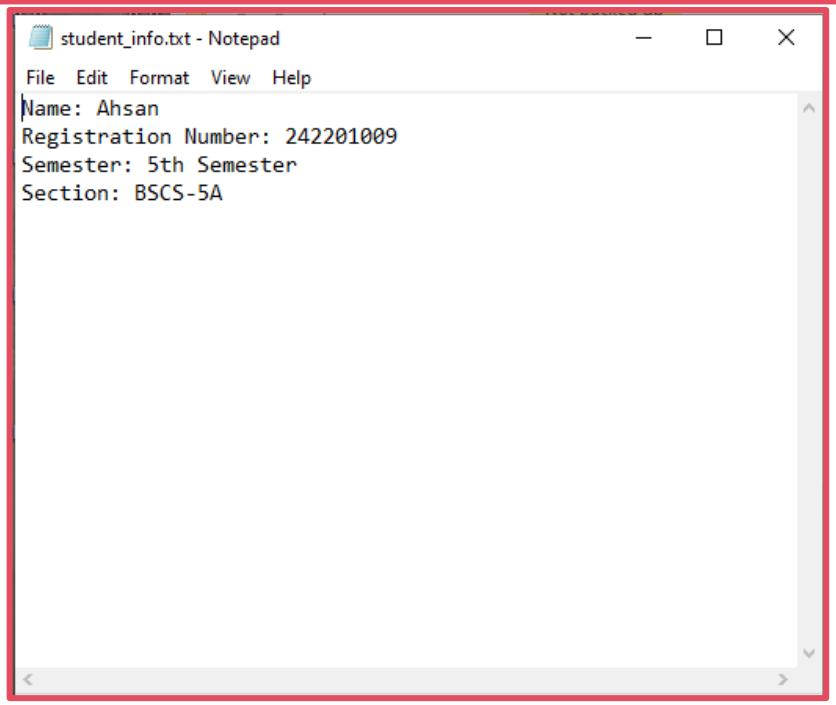
```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>echo Name: Ahsan > student_info.txt

C:\Users\Ahsan\Desktop\OS_Lab_Assignment>echo Registration Number: 242201009 >> student_info.txt

C:\Users\Ahsan\Desktop\OS_Lab_Assignment>echo Semester: 5th Semester >> student_info.txt

C:\Users\Ahsan\Desktop\OS_Lab_Assignment>echo Section: BSCS-5A >> student_info.txt

C:\Users\Ahsan\Desktop\OS_Lab_Assignment>
```



- **Purpose:** The echo command writes text into a file. The > operator creates or overwrites a file, while >> appends additional text without deleting existing content.

Step 7: Display the File Contents

- **Command:** ***type student_info.txt***

```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>type student_info.txt
Name: Ahsan
Registration Number: 242201009
Semester: 5th Semester
Section: BSCS-5A
```

- **Purpose:** Displays the contents of the specified text file in Command Prompt.

Step 8: Copy the File into the Backup Folder

- **Command:** ***copy student_info.txt Backup***
- **Purpose:** Copies the student information file into the Backup directory.

```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>copy student_info.txt Backup\
        1 file(s) copied.
```

Step 9: Rename the Copied File

- **Command:** ***ren Backup\student_info.txt student_info_backup.txt***

```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>ren Backup\student_info.txt student_info_backup.txt
```

- **Purpose:** Renames the copied file without changing its original location.

Step 10: Display All Files and Folders

- **Command:** ***dir /s***

```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>dir /s
 Volume in drive C is os
 Volume Serial Number is BE2A-8502

 Directory of C:\Users\Ahsan\Desktop\OS_Lab_Assignment

30/10/2026  06:37 pm    <DIR>          .
30/10/2026  06:37 pm    <DIR>          ..
30/10/2026  06:46 pm    <DIR>          Backup
30/10/2026  06:29 pm    <DIR>          Reports
30/10/2026  06:29 pm    <DIR>          Scripts
30/10/2026  06:29 pm    <DIR>          Students
30/10/2026  06:39 pm                91 student_info.txt
               1 File(s)             91 bytes

 Directory of C:\Users\Ahsan\Desktop\OS_Lab_Assignment\Backup

30/10/2026  06:46 pm    <DIR>          .
30/10/2026  06:46 pm    <DIR>          ..
30/10/2026  06:39 pm                91 student_info_backup.txt
               1 File(s)             91 bytes

 Directory of C:\Users\Ahsan\Desktop\OS_Lab_Assignment\Reports

30/10/2026  06:29 pm    <DIR>          .
30/10/2026  06:29 pm    <DIR>          ..
30/10/2026  06:29 pm    <DIR>          Final
30/10/2026  06:29 pm    <DIR>          Weekly
               0 File(s)              0 bytes

 Directory of C:\Users\Ahsan\Desktop\OS_Lab_Assignment\Reports\Final

30/10/2026  06:29 pm    <DIR>          .
30/10/2026  06:29 pm    <DIR>          ..
               0 File(s)              0 bytes
```

- **Purpose:** Displays all files and subdirectories, including their contents, within the current directory and its subdirectories.

Result: I successfully created the required directory structure using Command Prompt. I also created a student information file, displayed its contents, copied it into the Backup folder, and renamed the copied file.

Task 3 – CMD System Investigation

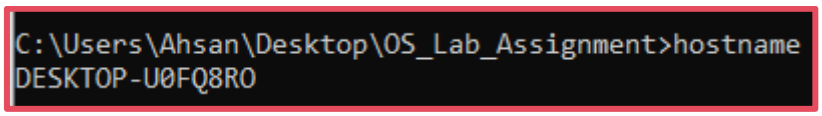
Objective

To investigate the computer system using Command Prompt and collect important system, user, network, process, and date/time information.

As a Junior System Administrator, it is important to know how to identify computer details, check network connectivity, and monitor running processes.

1. Computer Name

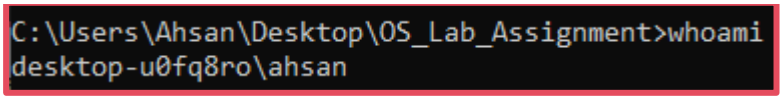
- **Command:** ***hostname***
- **Purpose:** Displays the name assigned to the computer.
- **Output:** Displays the computer's hostname.



- **Explanation:** This command helps identify a computer on a network and distinguish it from other computers.

2. Current Username

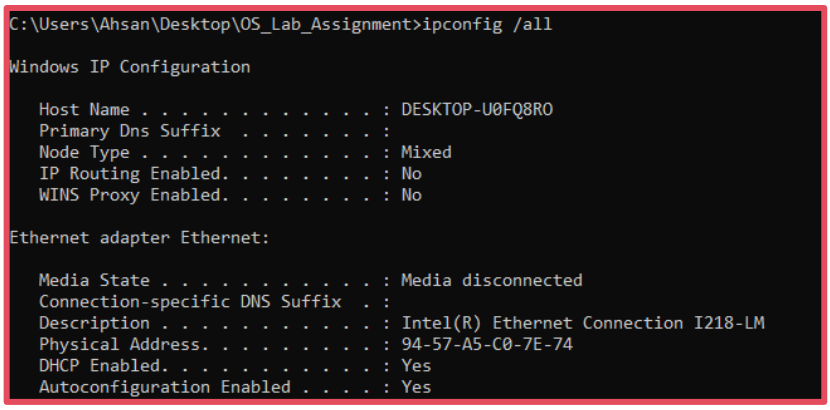
- **Command:** ***whoami***
- **Purpose:** Displays the username of the currently logged-in Windows user.
- **Output:** Displays the current user's account name, usually in the format ComputerName\Username or Domain\Username.



- **Explanation:** This command helps identify which user account is currently accessing the system.

3. Windows and IP Configuration

- **Command:** ***ipconfig /all***
- **Purpose:** Displays detailed network configuration information for all network adapters.
- **Output:**



```
Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 10-4A-7D-EF-59-C1
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 11:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #3
   Physical Address. . . . . . . . . : 12-4A-7D-EF-59-C0
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Dual Band Wireless-N 7260
   Physical Address. . . . . . . . . : 10-4A-7D-EF-59-C0
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::cad3:d0c2:5ab5:945b%16(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.100.5(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Friday, 30 October 2026 6:02:47 pm
   Lease Expires . . . . . . . . . . : Saturday, 31 October 2026 6:02:46 pm
   Default Gateway . . . . . . . . . : 192.168.100.1
   DHCP Server . . . . . . . . . . . : 192.168.100.1
   DHCPv6 IAID . . . . . . . . . . . : 437275261
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-05-12-7B-94-57-A5-C0-7E-74
   DNS Servers . . . . . . . . . . . : 192.168.100.1
   NetBIOS over Tcpip. . . . . . . . : Enabled

C:\Users\Ahsan\Desktop\OS_Lab_Assignment>
```

- **Explanation:** This command is useful for investigating network settings and troubleshooting connectivity problems.

4. Current IP Address

- **Command:** ***ipconfig***
- **Purpose:** Displays the basic IP configuration of active network adapters.
- **Output:** Shows the IPv4 address, subnet mask, and default gateway where available.

```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 11:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

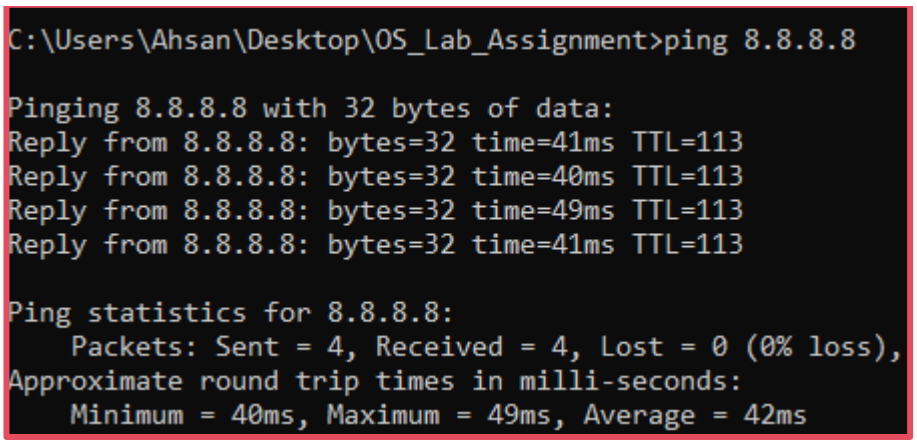
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::cad3:d0c2:5ab5:945b%16
   IPv4 Address. . . . . . . . . . . : 192.168.100.5
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.100.1
```

- **Explanation:** An IP address identifies a device on an IP network and allows it to communicate with other devices.

5. Network Connectivity Status

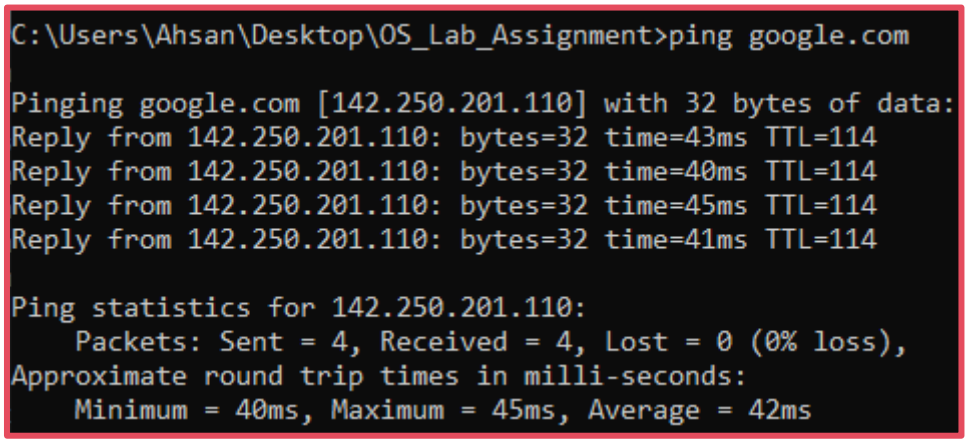
- **Command: ping 8.8.8.8**
- **Purpose:** Tests network connectivity by sending packets to Google's public DNS server.
- **Output:** If successful, it displays replies from the destination. If unsuccessful, it may display a timeout or destination unreachable message.



- **Explanation:** This command helps determine whether the computer can communicate with an external IP address.

Additional DNS Test: ***ping google.com***

- **Purpose:** Tests connectivity using a domain name and also checks whether DNS name resolution is working.



- **Explanation:** If the IP address ping works but the domain name ping fails, a DNS resolution problem may exist.

6. Running Processes

- **Command:** ***tasklist***
- **Purpose:** Displays a list of currently running processes on the computer.
- **Output:** Shows process names, process IDs (PID), session information, and memory usage.

```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>tasklist

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
System Idle Process              0 Services                   0          8 K
System                           4 Services                   0        136 K
Secure System                   56 Services                   0     22,348 K
Registry                       104 Services                   0     32,484 K
smss.exe                       428 Services                   0        984 K
csrss.exe                      596 Services                   0      5,012 K
wininit.exe                    776 Services                   0      5,324 K
services.exe                   852 Services                   0      9,568 K
LsaIso.exe                     864 Services                   0      1,824 K
lsass.exe                      884 Services                   0     22,452 K
svchost.exe                   1016 Services                   0     30,796 K
fontdrvhost.exe                320 Services                   0      2,112 K
svchost.exe                    668 Services                   0     22,124 K
svchost.exe                    736 Services                   0      9,524 K
svchost.exe                   1140 Services                   0      3,444 K
svchost.exe                   1168 Services                   0      9,400 K
svchost.exe                   1192 Services                   0      7,100 K
svchost.exe                   1256 Services                   0      5,880 K
svchost.exe                   1312 Services                   0     13,360 K
svchost.exe                   1544 Services                   0      4,892 K
svchost.exe                   1624 Services                   0     12,536 K
svchost.exe                   1636 Services                   0      5,516 K
svchost.exe                   1688 Services                   0      3,540 K
svchost.exe                   1696 Services                   0      8,692 K
svchost.exe                   1764 Services                   0     17,076 K
svchost.exe                   1800 Services                   0      8,996 K
wsc_proxy.exe                 1864 Services                   0      6,008 K
svchost.exe                   1904 Services                   0      5,088 K
svchost.exe                   1936 Services                   0      9,232 K
svchost.exe                   1952 Services                   0      6,828 K
svchost.exe                   1984 Services                   0      2,132 K
svchost.exe                   2032 Services                   0      8,668 K
svchost.exe                    944 Services                   0      3,688 K
Memory Compression            2136 Services                   0    460,532 K
svchost.exe                   2184 Services                   0      4,916 K
igfxCUIService.exe            2264 Services                   0      5,120 K
```

- **Explanation:** This command helps a system administrator monitor active applications and background processes.

7. Current Date

- **Command:** ***date /t***
- **Purpose:** Displays the current system date without asking the user to change it.
- **Output:** Shows the current date according to the computer's regional settings.

```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>date /t
30/10/2026
```

8. Current Time

- **Command:** ***time /t***
- **Purpose:** Displays the current system time without prompting for a new value.
- **Output:** Shows the current time according to the computer's regional settings.

```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>time /t
07:05 pm
```

Result: I successfully investigated the Windows computer using CMD commands. I identified the computer name, current username, network configuration, IP address, connectivity status, running processes, and current date and time.

Task 4 – Batch Script Automation

Objective

To create and execute a batch script that automatically performs routine system administration tasks and reduces the need for manual command execution.

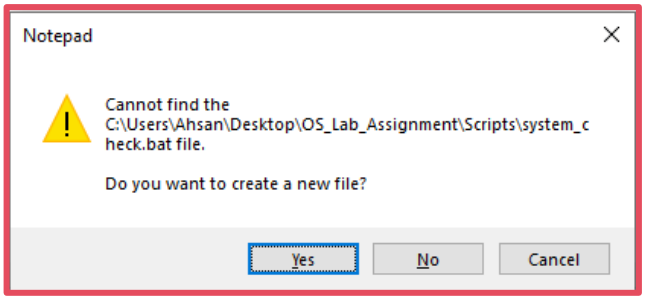
Step 1: Create the Batch File

Opening Command Prompt and making sure the current directory is: OS_Lab_Assignment

Opening Notepad using:
notepad Scripts\system_check.bat

```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>notepad Scripts\system_check.bat
```

If Windows asks to create a new file, select Yes.



Step 2: Write the Batch Script

Copy the following code into Notepad:

system_check.bat - Notepad

File Edit Format View Help

```
@echo off
title Windows System Check
color 0A
echo ======================================
echo       SYSTEM ADMINISTRATION TOOL
echo ======================================

echo.
echo Welcome to the System Check Program!

echo.
echo Current Username:
whoami
echo.
echo Computer Name:
hostname
echo.
echo Current Date:
date /t

echo.
echo Current Time:
time /t

echo.
echo Network Information:
ipconfig

echo.
echo Creating Daily Backup Folder...
set "BASE=%~dp0.."
set "BACKUP=%BASE%\Daily_Backup"
if not exist "%BACKUP%" mkdir "%BACKUP%"
echo.
echo Copying Student Information File...
if exist "%BASE%\student_info.txt" (
    copy /Y "%BASE%\student_info.txt" "%BACKUP%\"
    echo Student information file copied successfully.
) else (
    echo Error: student_info.txt was not found.
)
echo.
echo ======================================
echo SYSTEM CHECK COMPLETED SUCCESSFULLY
echo ======================================
pause
```

Saving the file with the following name:
system_check.bat

Step 3: Execute the Batch Script

Returning to Command Prompt and execute:
Scripts\system_check.bat

The batch file will automatically:

- Display a welcome message.
- Display the current username.
- Display the computer name.
- Display the current date and time.
- Display network configuration.
- Create the Daily_Backup folder if it does not exist.
- Copy student_info.txt into the backup folder.
- Display a completion message.
- Pause execution until a key is pressed.

```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>notepad Scripts\system_check.bat

C:\Users\Ahsan\Desktop\OS_Lab_Assignment>Scripts\system_check.bat
=====================================
      SYSTEM ADMINISTRATION TOOL
=====================================

Welcome to the System Check Program!

Current Username:
desktop-u0fq8ro\ahsan

Computer Name:
DESKTOP-U0FQ8RO

Current Date:
30/10/2026

Current Time:
07:14 pm

Network Information:

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 11:
```

Step 4: Verify the Backup Folder

- **Execute:** ***dir Daily_Backup***
- **Purpose:** Displays the contents of the Daily_Backup folder to verify that the student information file has been copied successfully.

```
C:\Users\Ahsan\Desktop\OS_Lab_Assignment>dir Daily_Backup
 Volume in drive C is os
 Volume Serial Number is BE2A-8502

 Directory of C:\Users\Ahsan\Desktop\OS_Lab_Assignment\Daily_Backup

30/10/2026  07:14 pm    <DIR>          .
30/10/2026  07:14 pm    <DIR>          ..
30/10/2026  06:39 pm                91 student_info.txt
               1 File(s)             91 bytes
               2 Dir(s)   1,945,862,144 bytes free
```

Explanation of Important Batch Commands

Command	Explanation
@echo off	Prevents commands from being displayed during script execution.
title	Sets the title of the Command Prompt window.
color 0A	Changes the console background and text color.
echo	Displays messages on the screen.
whoami	Displays the current username.
hostname	Displays the computer name.
date /t	Displays the current date.
time /t	Displays the current time.
ipconfig	Displays network configuration.
set	Creates environment variables for storing values.
if not exist	Checks whether a folder exists before creating it.
mkdir	Creates a new directory.
if exist	Checks whether the specified file exists.
copy /Y	Copies a file and automatically overwrites an existing destination file without asking.
pause	Stops the script temporarily and waits for a key press.

Result: I successfully created and executed a batch script named system_check.bat. The script automatically collected basic system information, created a backup directory, and copied the student information file into it.

This demonstrates how batch scripting can automate repetitive administrative tasks and save time.

Task 5 – PowerShell Administration

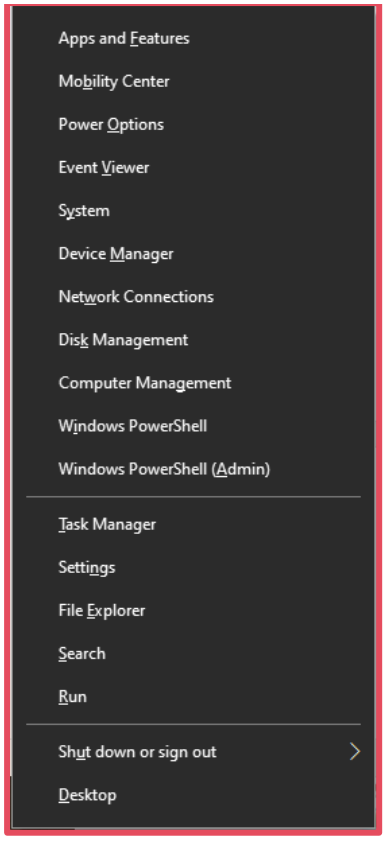
Objective

To perform basic Windows administration tasks using PowerShell, including directory management, file creation, process monitoring, and system information retrieval.

PowerShell is a powerful command-line shell and scripting language designed for system administration and automation.

Step 1: Open PowerShell

Press Win + X and select Windows PowerShell or Windows PowerShell (Admin) if available.



Step 2: Navigate to the Assignment Directory

- **Execute:** ***Set-Location "$HOME\Desktop\OS_Lab_Assignment"***

```
PS C:\Users\Ahsan> Set-Location "$HOME\Desktop\OS_Lab_Assignment"
PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment>
```

- **Purpose:** Changes the current working directory to the assignment folder on the Desktop.

Step 3: Display the Current Working Directory

- **Command:** ***Get-Location***

```
PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment> Get-Location

Path
----
C:\Users\Ahsan\Desktop\OS_Lab_Assignment
```

- **Purpose:** Displays the current working directory.
- **Explanation:** It helps identify the location where PowerShell commands will operate.

Step 4: List Files and Folders

- **Command:** ***Get-ChildItem***

```
PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment> Get-ChildItem


    Directory: C:\Users\Ahsan\Desktop\OS_Lab_Assignment


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/30/2026   6:46 PM                Backup
d-----        10/30/2026   7:14 PM                Daily_Backup
d-----        10/30/2026   6:29 PM                Reports
d-----        10/30/2026   7:09 PM                Scripts
d-----        10/30/2026   6:29 PM                Students
-a----        10/30/2026   6:39 PM             91 student_info.txt


PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment>
```

- **Purpose:** Displays all files and folders present in the current directory.
- **Explanation:** This command is similar to the dir command in CMD.

Step 5: Create a New Folder

- **Command:** ***New-Item -Path "PowerShell_Task" -ItemType Directory***

```
PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment> New-Item -Path "PowerShell_Task" -ItemType Directory


    Directory: C:\Users\Ahsan\Desktop\OS_Lab_Assignment


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/30/2026   7:28 PM                PowerShell_Task
```

- **Purpose:** Creates a new directory named PowerShell_Task.

Step 6: Create a Text File

- **Command:** ***New-Item -Path "PowerShell_Task\student_details.txt" -ItemType File***

```
PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment> New-Item -Path "PowerShell_Task\student_details.txt" -ItemType File


    Directory: C:\Users\Ahsan\Desktop\OS_Lab_Assignment\PowerShell_Task


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        10/30/2026   7:30 PM              0 student_details.txt
```

- **Purpose:** Creates a new text file named student_details.txt inside the PowerShell_Task folder.

Step 7: Write Student Information into the File

- **Command:** ***Set-Content -Path "PowerShell_Task\student_details.txt" -Value "Name: Ahsan"***
- **Append the registration number:*****Add-Content -Path "PowerShell_Task\student_details.txt" -Value "Registration Number: 242201009"***

```
PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment> Set-Content -Path "PowerShell_Task\student_details.txt" -Value "Name: Ahsan"
PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment> Add-Content -Path "PowerShell_Task\student_details.txt" -Value "Registration Number: 242201009"
```

- **Purpose:** Set-Content writes information into a file, while Add-Content adds additional information without removing the existing content.

Step 8: Read and Display File Contents

- **Command:** ***Get-Content -Path "PowerShell_Task\student_details.txt"***
- **Purpose:** Reads and displays the contents of the specified text file in PowerShell.
- **Output:**

```
PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment> Get-Content -Path "PowerShell_Task\student_details.txt"
Name: Ahsan
Registration Number: 242201009
PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment>
```

Step 9: Display Currently Running Processes

- **Command:** ***Get-Process***

```
PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment> Get-Process

Handles  NPM(K)    PM(K)      WS(K)     CPU(s)      Id  SI ProcessName
-------  ------    -----      -----     ------      --  -- -----------
    758      29    16860      16280               6400   0 afwServ
    147       8     3240       5364               9224   0 AggregatorHost
    231      20     3984       4724               6528   0 AppleMobileDeviceServ
ice
    340      20    11172      27792       1.53   25868   4 ApplicationFrameHost
    249      25    40536      20776              13984   0 aswEngSrv
   1108      31    68800      61976               5988   0 aswidsagent
   2032      55    49976      39864               3520   0 aswToolsSvc
    604      18    11264      19332      18.77   20860   0 audiodg
   7307     124   363724     213212               3248   0 AvastSvc
    699      33    23532      39632       1.63    2248   4 AvastUI
    518      28    15768      30756       0.73   19444   4 AvastUI
    578      31    17848      38876       2.14   23796   4 AvastUI
    552      31    16232      41272       0.42   24700   4 AvastUI
   3008      59    62384      80560     185.33   24964   4 AvastUI
    534      36   179296     115456      15.64     820   4 chrome
    472      53   603600     546044     610.55    2084   4 chrome
    394      51   280492     229768      56.09    6948   4 chrome
    348      37   258732     223088      40.28    7200   4 chrome
    261      24    42416      51544       7.36    9136   4 chrome
```

- **Purpose:** Displays all currently running processes on the computer.

- **Explanation:** It provides information such as process names, IDs, CPU usage, and memory usage.

Step 10: Display Basic System Information

- **Command:** ***Get-ComputerInfo | Select-Object WindowsProductName, WindowsVersion, OsArchitecture, CsName, CsManufacturer, CsModel***
- **Purpose:** Retrieves important information about the Windows operating system and computer hardware.
- **Output:** Displays:

  - Windows Product Name
  - Windows Version
  - Operating System Architecture
  - Computer Name
  - Computer Manufacturer
  - Computer Model

```
PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment> Get-ComputerInfo | Select-Object WindowsProductName, WindowsVersion, OsArchitecture, CsName, CsManufactu
rer, CsModel


WindowsProductName : Windows 10 Pro
WindowsVersion     : 2009
OsArchitecture     : 64-bit
CsName             : DESKTOP-U0FQ8RO
CsManufacturer     : Hewlett-Packard
CsModel            : HP EliteBook Folio 9480m
```

- **Explanation:** This command is useful for identifying the operating system and hardware details of a computer.

Step 11: Copy the PowerShell File into the Backup Folder

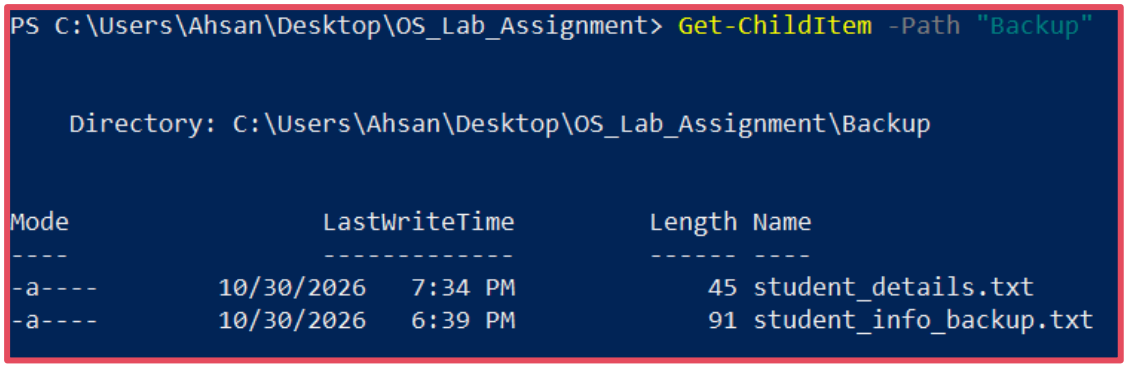
- **Command:** ***Copy-Item -Path "PowerShell_Task\student_details.txt" -Destination "Backup\"***

```
PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment> Copy-Item -Path "PowerShell_Task\student_details.txt" -Destination "Backup\"
PS C:\Users\Ahsan\Desktop\OS_Lab_Assignment>
```

- **Purpose:** Copies the PowerShell-created student details file into the existing Backup folder.
- **Explanation:** Copy-Item is used to copy files and directories from one location to another.

Step 12: Verify the Backup File

- **Command:** ***Get-ChildItem -Path "Backup"***



- **Purpose:** Displays the contents of the Backup folder to verify that the file was copied successfully.

Result: I successfully performed file and directory management using PowerShell. I created a new folder and text file, wrote and displayed student information, monitored running processes, retrieved computer information, and copied the file into the Backup directory.

Task 6 – CMD vs PowerShell Comparison

Objective

To compare Command Prompt and PowerShell based on their command style, ease of use, automation capabilities, and system administration features.

Feature	Command Prompt (CMD)	PowerShell
Command Style	Uses traditional text-based commands.	Uses commandlets (cmdlets) with a Verb-Noun naming structure.
Ease of Use	Simple and suitable for basic tasks.	Requires some learning but provides more advanced functionality.
Automation	Supports automation through batch files (.bat).	Supports advanced automation through PowerShell scripts (.ps1).
System Administration	Provides basic system management and troubleshooting commands.	Provides advanced Windows administration and system management capabilities.
Data Handling	Primarily works with text-based command output.	Works with structured objects and their properties.

Practical Comparison:

During this assignment, I used both Command Prompt and PowerShell to perform different system administration tasks. CMD uses traditional commands such as mkdir, copy, dir, and tasklist. It is simple to understand

and useful for basic file management, network troubleshooting, and executing batch scripts.

PowerShell uses a different command structure based on cmdlets, such as Get-Process, Get-Location, New-Item, and Copy-Item. It is more powerful because it works with structured objects instead of only plain text. This makes it easier to filter, organize, and process system information.

Both tools support automation, but CMD mainly uses batch files, whereas PowerShell supports more advanced scripting and automation. PowerShell also provides better integration with Windows system administration features.

Based on my practical experience, I would prefer PowerShell for advanced Windows administration because it offers greater flexibility, powerful scripting capabilities, and better system management features. However, CMD remains useful for quick and simple administrative operations.

Conclusion

In this assignment, I learned how to perform basic Windows system administration tasks using keyboard shortcuts, Command Prompt, Batch Scripting, and PowerShell.

I practically created and managed directories, generated text files, copied and renamed files, investigated system information, checked network connectivity, and monitored running processes.

I also developed and executed a batch script to automate routine administrative operations, including creating a backup folder and copying student information.

Furthermore, I explored PowerShell commands for file management, process monitoring, and retrieving computer information.

Overall, this assignment improved my practical understanding of Windows operating system administration and demonstrated how command-line tools and scripting can make routine administrative tasks easier, faster, and more efficient.